

Jaehong Kim

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Research Interest

AI for systems, AI for video streaming, Immersive video, Systems for large-scale AI, Networked system

Work Experience

Inha University

ASSISTANT PROFESSOR, DEPARTMENT OF ARTIFICIAL INTELLIGENCE

- Joint Appointment in Department of Electrical and Computer Engineering

Incheon, S.Korea

Sep. 2025 - Present

Carnegie Mellon University

POSTDOCTORAL RESEARCHER, COMPUTER SCIENCE DEPARTMENT

- Postdoctoral Fellowship Program granted by NRF. (Advisor: Srinivasan Seshan and Anthony Rowe)

Pittsburgh, PA, USA

Sep. 2024 - Aug. 2025

Education

KAIST (Korea Advanced Institute of Science and Technology)

PH.D. IN ELECTRICAL ENGINEERING

- Thesis title: Enabling High-quality 2D and 3D Live Streaming at Ingest (Advisor: Prof. Dongsu Han)

Daejeon, S.Korea

Feb. 2020 - Aug. 2024

KAIST (Korea Advanced Institute of Science and Technology)

M.S. IN ELECTRICAL ENGINEERING

- Thesis title: Enhancing Live Video Quality at Ingest Using Online Trained DNNs (Advisor: Prof. Dongsu Han)

Daejeon, S.Korea

Sep. 2018 - Feb. 2020

KAIST (Korea Advanced Institute of Science and Technology)

B.S. IN ELECTRICAL ENGINEERING (CUM LAUDE)

Daejeon, S.Korea

Mar. 2014 - Aug. 2018

University of Maryland

EXCHANGE STUDENT PROGRAM

College Park, MD, USA

Jan. 2016 - May. 2016

Publications / Preprints

CONFERENCE PROCEEDINGS (C), JOURNAL(J), WORKSHOPS (W), PREPRINTS (P)

TOPICS

[C-10] FlaME: MAC Scheduling at Scale in Virtualized Radio Access Network

5G/6G, vRAN

Yunheon Lee*, Jaehong Kim[†], and Dongsu Han[†]

USENIX NSDI 2027, * Co-first, [†] Co-corresponding

[C-9] BlenDR: Bandwidth-efficient RGB-D Representation and Delivery for Live 3D Video Streaming

Volumetric Video

Jaehong Kim, Joon Ha Kim, Yunheon Lee, and Dongsu Han

ACM MobiSys 2026, 🌐 webpage, 📺 Demo

[C-8] Neural-Centric Video Processing Pipeline for Unified Multi-Task Inference

AI for Video Systems

Seyeon Lee, Juncheol Ye, Jaehong Kim, and Dongsu Han

IEEE/CVF CVPR 2026

[C-7] SceneHub4D: A Dataset and Evaluation Framework for 6-DoF 4D VR Scenes

Immersive Media

Jaehong Kim, Tao Jin, Mallesham Dasari, Srinivasan Seshan, and Anthony Rowe

IEEE VR 2026 / Top submission, accepted to IEEE TVCG, 🌐 webpage

[C-6] NerVast: Compression-efficient Scaling of Implicit Neural Video Representations via

AI for Video

Scene-based Parameter-sharing

Yunheon Lee, Juncheol Ye, Jaehong Kim[†], and Dongsu Han[†]

IEEE/CVF WACV 2026, [†] Co-corresponding

[C-5] FlexPass: A Case for Flexible Credit-based Transport for Datacenter Networks

Datacenter Networking

Hwijoon Lim, Jaehong Kim, Inho Cho, Keon Jang, Wei Bai, and Dongsu Han

ACM EuroSys 2023, 🌐 webpage

[C-4] OutRAN: Co-optimizing for Flow Completion Time in Radio Access Network

5G Networks

Jaehong Kim, Yunheon Lee, Hwijoon Lim, Youngmok Jung, Song Min Kim, and Dongsu Han

ACM CoNEXT 2022 (Best Paper Finalist), 🌐 webpage

[C-3] NeuroScaler: Neural Video Enhancement at Scale

AI for Live Streaming

Hyunho Yeo, Hwijoon Lim, Jaehong Kim, Youngmok Jung, Juncheol Ye, and Dongsu Han

ACM SIGCOMM 2022, 🌐 webpage

[C-2] Neural-Enhanced Live Streaming: Improving Live Video Ingest via Online Learning Jaehong Kim*, Youngmok Jung*, Hyunho Yeo, Juncheol Ye, and Dongsu Han ACM SIGCOMM 2020 , * Co-first authors, webpage	AI for Live Streaming
[C-1] Neural Adaptive Content-aware Internet Video Delivery Hyunho Yeo, Youngmok Jung, Jaehong Kim, Jinwoo Shin, and Dongsu Han USENIX OSDI 2018 , webpage , DASH.js code	AI for Video Streaming
[J-1] Efficient Disaggregated Cloud Storage for Cold Videos with Neural Enhancement Jinyeong Lim, Juncheol Ye, Jaehong Kim, Hwijoon Lim, Hyunho Yeo, Junhyeok Jang, Myoungsoo Jung, and Dongsu Han IEEE Micro	AI for Cloud Storage
[W-3] Reconstructing Reality over Time: From Drone Capture to Timelapse Gaussian Splatting Jaehong Kim, Srinivasan Seshan, and Anthony Rowe IEEE ISMAR 2025 (Poster)	Immersive Media
[W-2] Presto: Hybrid CPU-GPU Preprocessing Framework for Video-based AI Inference System Ji Hyuk Lee, Dongsu Han, and Jaehong Kim ACM NetAISys 2025	System for AI Inference
[W-1] Neural Cloud Storage: Innovative Cloud Storage Solution for Cold Video Jinyeong Lim, Juncheol Ye, Jaehong Kim, Hwijoon Lim, Hyunho Yeo, and Dongsu Han ACM HotStorage 2023 , webpage	AI for Cloud Storage

Honors and Awards

Jun. 2026	NVIDIA Academic Grant Program Selected for NVIDIA Academic Grant Program for the project “Spatial Intelligence Platform for Large-Scale Time-Varying Digital Twins”; awarded 4× RTX PRO 6000 Max-Q Workstation Edition GPUs.	<i>NVIDIA</i>
Mar. 2026	Outstanding Young Scientist Grant - National Research Foundation of Korea Awarded for a three-year project on building a large-scale digital twin system; total funding: ₩300,000,000.	<i>NRF</i>
Mar. 2025	NSF Workshop on Networking and Systems Challenges in Immersive Computing — Travel Grant Position paper accepted for the NSF Workshop with travel grant support.	<i>NSF</i>
Jan. 2025	NSF NeTS Early Career Workshop with Travel Grant Selected to attend the NeTS Early Career Workshop 2025 at NSF Headquarters.	<i>NSF</i>
Sep. 2024	NRF Postdoctoral Fellowship Program Selected as a principal investigator of Postdoctoral Fellowship Program (Nurturing Next-generation Researchers) in 2024 granted by the National Research Foundation of Korea (NRF) with ₩60,000,000 grant for one year.	<i>NRF</i>
Feb. 2023	29th Samsung Humantech Paper Award Awarded Silver-prize (118 out of 1972 papers), Communication & Network	<i>Samsung Electronics</i>
Dec. 2022	Google Conference Scholarship Travel grants for students giving oral presentations at top-tier CS conferences.	<i>Google LLC</i>
Dec. 2022	ACM CoNEXT’22 Best Paper Award Nomination & ACM Student Grant Received the highest review score with five “4 Accept” ratings.	<i>NSF & ACM</i>
Feb. 2022	28th Samsung Humantech Paper Award Gold Prize (1st place), Communication & Network	<i>Samsung Electronics</i>
2021	KAIST Breakthrough of the Year For the top 15 most significant research achievements.	<i>KAIST</i>
2020	Donghwa Industry Moon Daewon AI Research Scholarship Awarded to a graduate student for outstanding AI research and collaborative spirit.	<i>KAIST</i>
2018	USENIX OSDI Student Grant	<i>USENIX</i>

Patents

INTERNATIONAL

US17265680	Live video ingest system and method	KAIST
US11463750B2	Method and apparatus for transmitting adaptive video in real time using content-aware neural network	KAIST
US20250240811A1 (Pending)	Method and device for scheduling in wireless communication system	Samsung Electronics & KAIST
DOMESTIC (SOUTH KOREA)		
KR10-2338986	Method for enhancing live video delivery at ingest point utilizing content-aware neural network	KAIST
KR10-2129115	Method and apparatus for transmitting adaptive video in real time using content-aware neural network	KAIST
KR10-2811422-0000	Neural cloud storage system and operating method thereof	KAIST
KR10-2024-0170218	Method of encoding and decoding video including depth data	KAIST
(Filed)	AI-native Media Processing Technology based on Neural Network Representation	KAIST
KR10-2023-0164365 (Filed)	Unified Compression Method for RGB and Depth Video in Live 3D Video Streaming	KAIST
KR10-2022-0091760 (Filed)	Acceleration method for encoding selective super-resolved video	KAIST
KR10-2022-0091726 (Filed)	Acceleration and scheduling method for video super-resolution based on codec-level information	KAIST
KR10-2022-0138553 (Filed)	Practical flow scheduling algorithm designed for 4G/5G radio access network base stations for low-latency applications	Samsung Electronics & KAIST
KR10-2022-0077669 (Filed)	Method of scheduling flow and electronic device performing the method	Samsung Electronics & KAIST

Mentoring Experience

Research Supervision at Inha University (2025–Present)

- Kangpyo Jeon (B.S. Inha Univ. / Jun. 2026 - Present): 3D Vision, Spatial Computing.
- Junghwan Lee (B.S. Inha Univ. / Jun. 2026 - Present): Efficient LLM/VLM Inference.
- Seono Choi (B.S. Inha Univ. / Jun. 2026 - Present): LLM/VLM Serving Systems.
- Bosung Seo (B.S. Inha Univ. / Dec.2025 - Present): Efficient Vision-Language Models (VLMs).
- Sangmin Park (B.S. Inha Univ. / Dec.2025 - Present): Multimodal RAG.
- Seoyeon Choi (B.S. Inha Univ. / Dec.2025 - Present): Agentic AI Systems.

Individual Study & Research Mentoring at KAIST

- JiHyuk Lee (B.S. CAU → Ph.D. Candidate at KAIST / Feb. 2025 - Present): Mentored research on video-based AI Inference system [W-3].
- Seyeon Lee (M.S. KAIST → Ph.D. Candidate at KAIST / Jun. 2024 - Present): Mentored research on Neural Video Pipeline [P-2].
- Joon Ha Kim (B.S. KAIST → Ph.D. Candidate at UT Austin / Jun. 2023 - Jan. 2024): Mentored research on live 3D streaming [P-2]. Read his experience 🏠 here.
- Yunheon Lee (B.S. KAIST → Ph.D. Candidate at KAIST / Jun. 2021 - Present): Mentoring research on 5G [C-4], and AI for video [P-1].
- Jinyeong Lim (M.S. KAIST → Ph.D. Candidate at KAIST): Mentored research on AI for cloud storage [W-1].
- Euijun Jeong (B.S. KAIST): Mentored research on an efficient cluster-wise training scheme for content-aware neural-enhancement.
- Hyojin Choi (B.S. KAIST / Jan.2023 - Jun.2023): Mentored research on deep neural video compression.

Teaching Experience

Instructor, Inha University

- Parallel Computing (AIE3006) Fall 2026
- Operating Systems (AIE3006) Spring 2026
- Algorithm (AIE3002) Spring 2025
- Computer Networks (AIE3009) Fall 2025, Fall 2026
- Object-oriented Programming 2 (AIE1002) Fall 2025

Teaching Assistant, KAIST

- Advanced Computer Networking and Cloud Computing (EE618)
- Network Programming (EE324)
- SK Hynix ASK Program
- Systems and Applications of Artificial Intelligence and Machine Learning (EE793)
- Programming Structures for Electrical Engineering (EE209)

Spring 2021
Fall 2020, Fall 2021
Aug. 2020
Spring 2020
Spring & Fall 2019, Spring & Fall 2022

Presentation

Conference talk at MobiSys 2026

Blender: Bandwidth-efficient RGB-D Representation and Delivery for Live 3D Video Streaming

Cambridge, UK

June. 2026

Conference talk at IEEE VR 2026

SceneHub4D: A Dataset and Evaluation Framework for 6-DoF 4D VR Scenes

Daegu, S. Korea

Mar. 2026

AI Convergence Seminar, Inha University

Media Technologies and Applications in the AI Era

Incheon, S. Korea

Oct. 2025

Research talk at Nokia Tech AI Visual Systems Research Team

Enabling High-quality 2D and 3D Live Streaming over the Internet

Vitruv, USA

April. 2025

Demo & Poster session at NSF Immercom'25 Workshop

Enabling 3D Live Streaming over the Internet

Arlington, VA, USA

April. 2025

Research talk at Qualcomm Immersive Video Research Team

Enabling High-quality 2D and 3D Live Streaming over the Internet

Vitruv, USA

Feb. 2025

Computer Science & Engineering Department Seminar at UNIST

Improving the Quality of Experience (QoE) of Internet Applications

Ulsan, S. Korea

Jun. 2024

Conference talk at CoNEXT'22

Presented OutRAN: Co-optimizing for Flow Completion Time in Radio Access Network.  Demo

Rome, Italy

Dec. 2022

Conference talk at SIGCOMM'20

Presented Neural-Enhanced Live Streaming: Improving Live Video Ingest via Online Learning.

Virtual

Aug. 2020

 20-min talk ,  10-min talk

Demo & Poster session at OSDI'18

Presented demo of Neural Adaptive Content-aware Internet Video Delivery.  Demo

Carlsbad, CA, USA

Oct. 2018

Academic Service

2023 - 2026	IEEE/ACM Transactions on Networking , Role: Reviewer
2026	IEEE ISMAR , Role: Reviewer
2025	IEEE VR , Role: Reviewer
2025	ACM Multimedia , Role: Reviewer
2025	USENIX Annual Technical Conference , Role: Reviewer and Paper Shepherd
2025	IEEE ICNP , Role: Session Chair

Skills

Programming Python, C/C++, JavaScript, CUDA

AI Frameworks TensorFlow, PyTorch, TensorRT

Other Skills dash.js, ffmpeg, NS-3, srsRAN, Docker

Languages Korean (native), English (fluent, IBT TOEFL 106)

References

Available upon request.